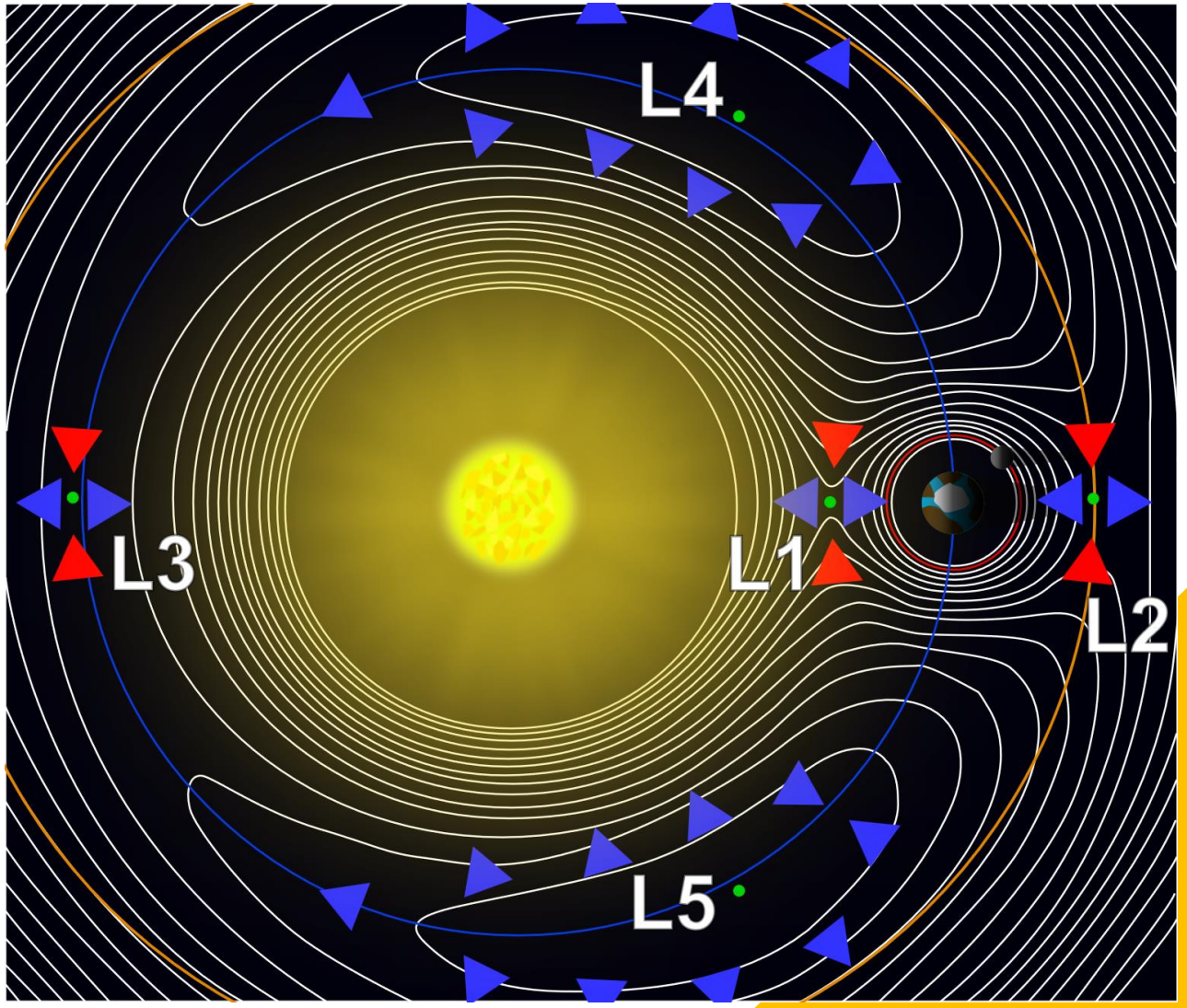
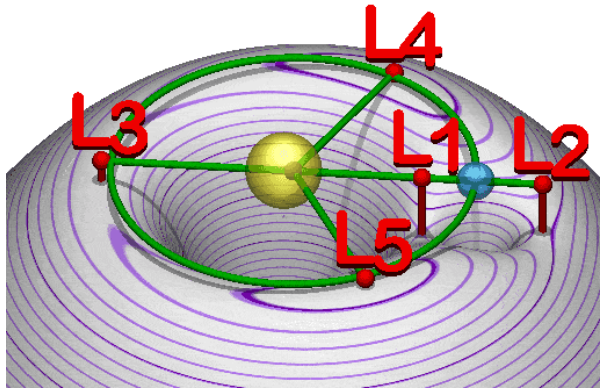
PHYSICS I
WEEK 10

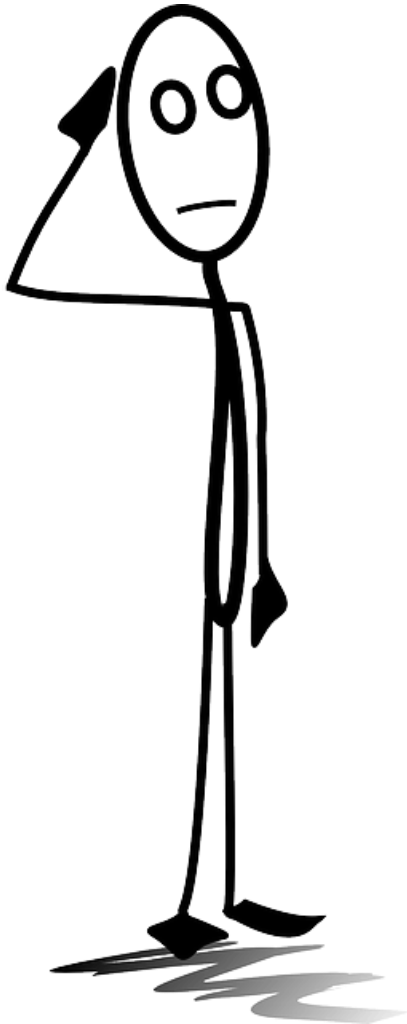
Siddharth Dhomkar
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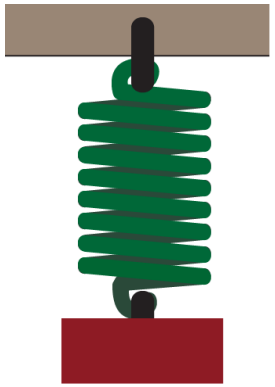
WEEK 10: OUTLINE

- Euler-Lagrange equations (Morin 6.1)
- Generalized coordinates and Lagrangian Dynamics (Morin 6.4, 6.5)





REVIEW: EULER-LAGRANGE EQUATIONS



Newton' law

$$F = ma$$

$$m\ddot{x} = -kx$$

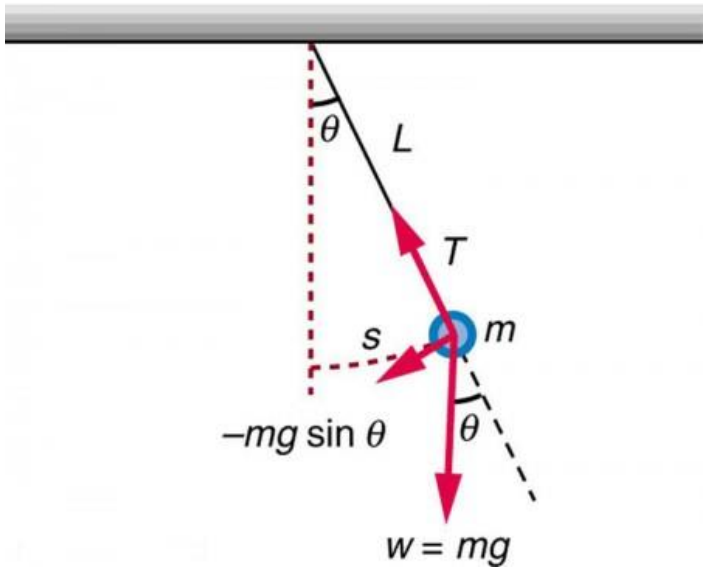
$$L = T - U$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

Euler-Lagrange equations

REVIEW: NEWTONIAN DYNAMICS

Simple pendulum



$$\boldsymbol{F} = F_r\hat{\boldsymbol{r}} + F_\theta\hat{\theta} = ma_r\hat{\boldsymbol{r}} + ma_\theta\hat{\theta}$$

$$T = mL\dot{\theta}^2 + mg\cos\theta$$

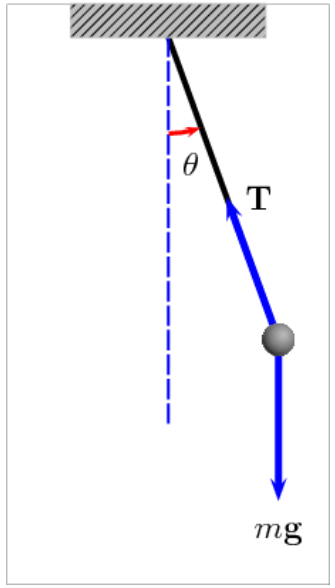
$$\ddot{\theta} = -\frac{g}{L}\sin\theta$$

REVIEW: EULER-LAGRANGE EQUATIONS

Simple pendulum

$$x = l \sin\theta$$

$$y = l \cos\theta$$



$$L = T - U$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

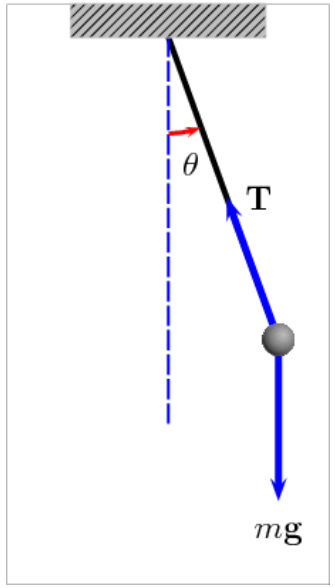
Euler-Lagrange equations

REVIEW: EULER-LAGRANGE EQUATIONS

Simple pendulum

$$x = l\sin\theta$$
$$y = l\cos\theta$$

$$T = \frac{1}{2}mv^2 \qquad U = -mgl\cos\theta$$



$$L = T - U$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

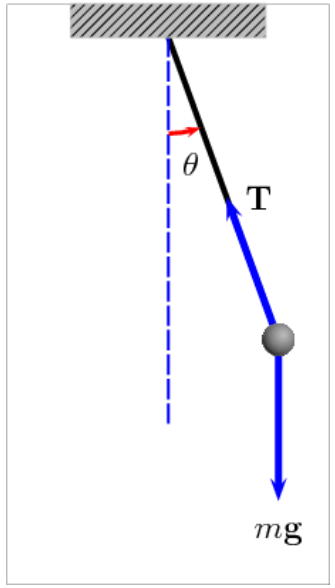
Euler-Lagrange equations

REVIEW: EULER-LAGRANGE EQUATIONS

Simple pendulum

$$x = l\sin\theta$$
$$y = l\cos\theta$$

$$T = \frac{1}{2}mv^2 \qquad U = -mgl\cos\theta$$



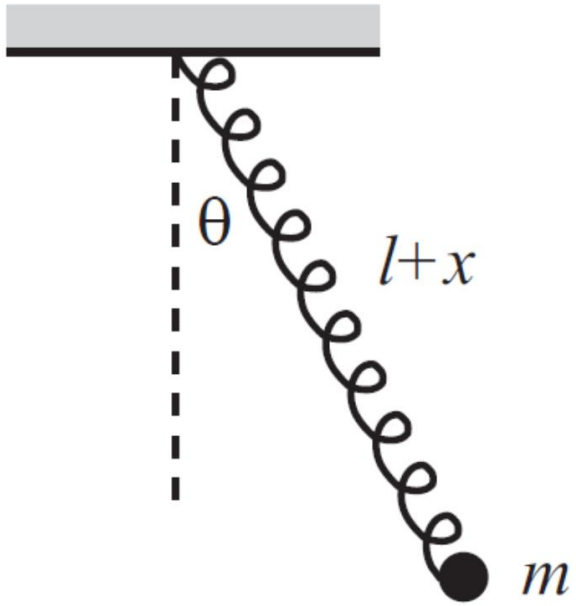
$$L = T - U$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

Euler-Lagrange equations

$$\ddot{\theta} = -\frac{g}{l}\sin\theta$$

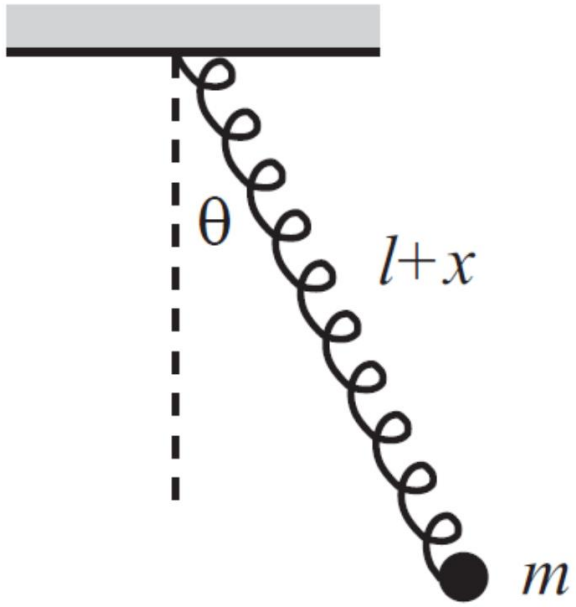
EULER-LAGRANGE EQUATIONS



$T =$

$V(x, \theta) =$

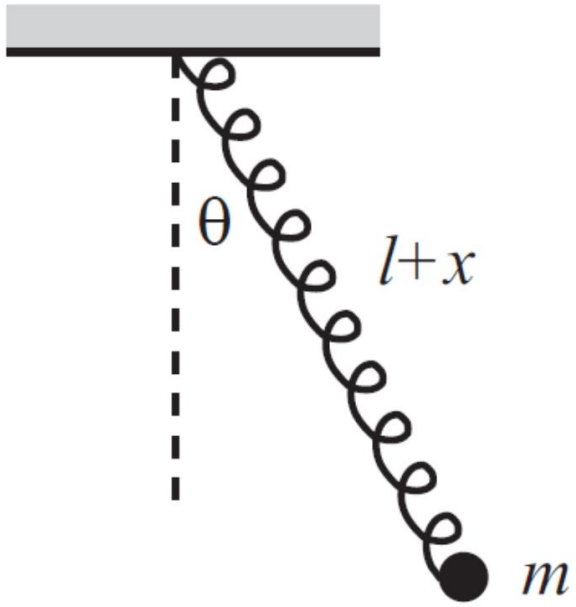
EULER-LAGRANGE EQUATIONS



$$T = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) \qquad V(x,\theta) = -mg(\ell + x)\cos\theta + \frac{1}{2}kx^2$$

$$L \equiv T - V = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) + mg(\ell + x)\cos\theta - \frac{1}{2}kx^2$$

EULER-LAGRANGE EQUATIONS



$$T = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) \qquad V(x,\theta) = -mg(\ell + x)\cos\theta + \frac{1}{2}kx^2$$

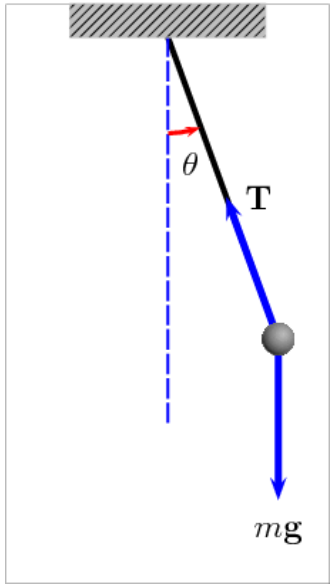
$$L \equiv T - V = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) + mg(\ell + x)\cos\theta - \frac{1}{2}kx^2$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x} \quad \Longrightarrow \quad m\ddot{x} = m(\ell + x)\dot{\theta}^2 + mg\cos\theta - kx$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \frac{\partial L}{\partial \theta} \quad \Longrightarrow \quad m(\ell + x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg\sin\theta$$

CONSERVATION LAWS

Simple pendulum – Take 3



$$r = l$$

$$q = \theta$$

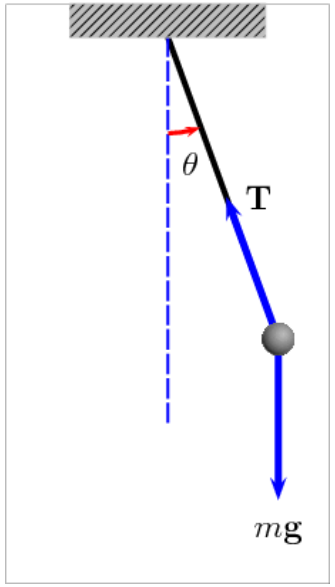
Generalized coordinates

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k}$$

CONSERVATION LAWS

Simple pendulum – Take 2



$$r = l$$

$$q = \theta$$

Generalized coordinates

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k}$$

- If Cartesian coordinates are used, then ${}^{\partial L}/_{\partial \dot{x}_k}$ is simply the momentum, $m\dot{x}_k$.
- ${}^{\partial L}/_{\partial \dot{q}_k}$ is called the **generalized momentum**.

CONSERVATION LAWS

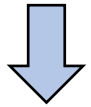
Cyclic coordinates

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k} = 0$$

CONSERVATION LAWS

Cyclic coordinates

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k} = 0$$



$$\frac{\partial L}{\partial \dot{q}_k} = C$$

Conserved Quantity

Example 1: Consider a ball thrown through the air.

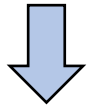
$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Conserved Quantity: Linear momentum

CONSERVATION LAWS

Cyclic coordinates

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k} = 0$$



$$\frac{\partial L}{\partial \dot{q}_k} = C$$

Conserved Quantity

Example 1: Consider a ball thrown through the air.

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Conserved Quantity: Linear momentum

Example 2: $V(r, \theta)$ in spherical coordinates.

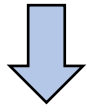
$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 sin^2\theta\dot{\emptyset}^2\right) - V(r, \theta)$$

Conserved Quantity: Angular momentum

CONSERVATION LAWS

Cyclic coordinates

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k} = 0$$



$$\frac{\partial L}{\partial \dot{q}_k} = C$$

Conserved Quantity

Example 1: Consider a ball thrown through the air.

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Conserved Quantity: Linear momentum

Example 2: $V(r, \theta)$ in spherical coordinates.

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 sin^2\theta\dot{\emptyset}^2\right) - V(r, \theta)$$

Conserved Quantity: Angular momentum

Example 3: $V(r)$ in cylindrical coordinates.

$$L = \frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\emptyset}^2 + \dot{z}^2\right) - V(r)$$

Conserved Quantity: Linear & angular momentum